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Paper 1: A Combined Experimental and Predictive Modeling Study of Diltiazem Release From

Bibliographic Information

- **Title:** A Combined Experimental and Predictive Modeling Study of Diltiazem Release From Natural Polymer Microcapsules in Different Media
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- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2026
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Quick Take

This paper demonstrates how in vitro release data can be integrated with mechanistic modeling in SimBiology to predict diltiazem release from microcapsules under varying pH conditions. The Weibull model outperformed traditional kinetic models, and VPC confirmed predictive performance. Pharmacometricians interested in formulation development, QbD, or in vitro-in vivo correlation will find the workflow directly applicable.

Executive Summary

This study integrates experimental in vitro release data of diltiazem-loaded microcapsules with mechanistic modeling to predict formulation performance under physiologically relevant conditions. The Weibull model provided the best fit in both distilled water and simulated saliva ($R^2 = 0.989$ and 0.994 , respectively), outperforming Higuchi and Korsmeyer-Peppas models based on AIC and BIC criteria. A two-compartment SimBiology model with first-order release kinetics was developed and validated using Visual Predictive Check (VPC) with $N = 200$ virtual individuals. The model was then extended to simulate the effect of sublingual pH variability (6.2–7.4), demonstrating that lower pH increases cumulative release due to a higher unionized fraction of diltiazem (pK_a 7.7). This work provides a scalable pharmacometric framework for transforming routine dissolution data into predictive models, though the internal validation and simplified kinetics warrant cautious interpretation.

Scientific Context & Motivation

Diltiazem is a first-line agent for acute rate control in atrial fibrillation, but no sublingual formulation exists, leaving a gap in pre-hospital management. Most in vitro release studies are descriptive and lack predictive power. This study addresses the need for biorelevant, predictive models that can simulate formulation performance under physiologically variable conditions (e.g., saliva pH). It challenges the traditional separation between formulation science and pharmacometrics by proposing an integrated experimental-modeling framework.

✂ Methodological Snapshot

The study employs a sequential modeling workflow: (1) kinetic model fitting (Higuchi, Korsmeyer-Peppas, Weibull) to experimental in vitro release data, with model selection based on AIC and BIC; (2) development of a two-compartment mechanistic model in SimBiology/MATLAB using first-order kinetics to estimate release rate constants; (3) internal validation via VPC with $N = 200$ virtual individuals; and (4) extension to a sublingual pH variability scenario (6.2–7.4) by scaling Weibull parameters according to the Henderson-Hasselbalch equation for diltiazem (pKa 7.7).

☒ Structural Model Breakdown

The mechanistic model is a two-compartment system in SimBiology. Compartment 1 represents the polymeric microcapsule matrix containing the encapsulated drug, initialized at $Q_0 = 19.31$ mg based on measured encapsulation efficiency. Compartment 2 represents the dissolution medium (distilled water or simulated saliva), initialized at zero. Drug transfer from compartment 1 to compartment 2 is governed by a first-order rate constant, $k_{release}$. The model does not include any other compartments or processes (e.g., degradation, absorption). The differential equation for the system is: $dQ_1/dt = -k_{release} \cdot Q_1$; $dQ_2/dt = k_{release} \cdot Q_1$. The Weibull model used for kinetic analysis and VPC is an empirical function: $F(t) = 1 - \exp[-(t/\lambda)^\beta]$, where $F(t)$ is the fractional release, λ is the scale parameter, and β is the shape parameter.

Detailed Methodological Analysis

Modeling Approach

Kinetic modeling using Higuchi, Korsmeyer-Peppas, and Weibull models. Mechanistic modeling using a two-compartment model in SimBiology/MATLAB: compartment 1 (capsule matrix, initial $Q_0 = 19.31$ mg) and compartment 2 (dissolution medium, initial 0 mg), with first-order transfer governed by $k_{release}$. Population simulations for pH scenario used the Weibull model with parameter scaling based on the unionized fraction.

Data Sources

In vitro release data from diltiazem-loaded gelatin B/gum arabic microcapsules in distilled water and simulated saliva at 37°C under sink conditions. Sampling at 12 time points from 5 to 720 min. Drug quantification by HPLC. Encapsulation efficiency ($Q_0 = 19.31$ mg) used to initialize the mechanistic model.

Estimation Methods

Kinetic model parameters (Higuchi, Korsmeyer-Peppas, Weibull) were estimated via nonlinear least squares regression. Mechanistic model parameters ($k_{release}$) were estimated using a regression-based optimization process in SimBiology/MATLAB, minimizing the discrepancy between experimental and simulated cumulative release profiles.

Model Evaluation

Model performance was assessed using R^2 , RMSE, AIC, and BIC for kinetic models. The mechanistic model was evaluated by comparing simulated vs. experimental profiles (R^2 , RMSE) and by Visual Predictive Check (VPC) with $N = 200$ virtual individuals, where Weibull parameters were sampled from log-normal distributions (CV 10% for Q_{max} and λ , 5% for β). All observed data fell within the 5th–95th percentile prediction interval.

Covariate Analysis

Covariate analysis was not performed in the traditional pharmacometric sense. Instead, a scenario analysis was conducted to evaluate the effect of saliva pH on release, using the Henderson-Hasselbalch equation to scale Weibull parameters (Q_{max} , λ) based on the unionized fraction of diltiazem relative to a reference pH of 6.8.

Statistical Rigor Assessment

The study uses appropriate information criteria (AIC, BIC) for model selection, which penalize overfitting. The VPC is a standard pharmacometric validation tool, though it is internal (same data for development and validation). The sample size for VPC ($N = 200$) is adequate. The pH scenario uses random sampling from a uniform distribution, which is reasonable for exploring variability. Limitations include the small number of experimental time points ($n = 12$) and the absence of an external validation dataset. Sensitivity analyses on the CV assumptions for VPC parameters were not reported.

Key Findings

The Weibull model provided the best fit for diltiazem release in both distilled water ($R^2 = 0.989$, AIC = -9.49) and simulated saliva ($R^2 = 0.994$, AIC = -19.36), outperforming Higuchi and Korsmeyer-Peppas models. The two-compartment mechanistic model estimated release rate constants of 0.01112 min^{-1} in water and 0.00263 min^{-1} in saliva, indicating significantly slower release in the biorelevant medium. VPC with $N = 200$ virtual individuals confirmed that all observed data fell within the 5th–95th percentile prediction interval. The pH variability scenario (6.2–7.4) showed that median cumulative release at 720 min ranged from 11.58 mg (pH 7.0–7.4) to 14.76 mg (pH 6.2–6.5), mechanistically linked to the pH-dependent unionized fraction of diltiazem (pKa 7.7).

Clinical & Regulatory Implications

The study provides a mechanistic rationale for developing a sublingual diltiazem formulation, addressing an important clinical gap in acute tachycardia management. The pH-dependent release simulations suggest that interindividual variability in saliva pH (6.2–7.4) could lead to meaningful differences in drug exposure, with lower pH favoring higher release. This implies that formulation optimization or patient stratification based on salivary pH might be necessary to ensure consis-

tent therapeutic effects. However, these are in silico predictions based on in vitro data; clinical pharmacokinetic studies are essential before any dosing recommendations can be made.

Strengths & Limitations

Strengths

- Clear integration of experimental data with mechanistic modeling and population simulation.
- Rigorous model selection using AIC and BIC, with consistent results across two media.
- Use of VPC for model validation, a standard in pharmacometrics but rare in formulation science.
- Clinically relevant pH scenario analysis that provides mechanistic insight into interindividual variability.
- Reproducible workflow using SimBiology/MATLAB with a fixed random seed.

Limitations (Acknowledged by Authors)

- Validation is internal; independent external datasets were not available.
- The first-order two-compartment model is a simplification; microcapsule release may involve polymer swelling, erosion, and anomalous diffusion.
- EDS results were used only as qualitative supportive information.
- Future studies should incorporate multi-mechanism models.

Limitations (Expert Review)

- The mechanistic model assumes first-order kinetics, which may not capture the initial burst or late-stage tailing often seen in microcapsule release.
- The VPC uses the same data for model development and validation, which can overestimate predictive performance.
- No formal uncertainty quantification (e.g., bootstrap or profile likelihood) was performed for the mechanistic model parameters.
- The pH scenario assumes a direct linear scaling of Weibull parameters with the unionized fraction, which may not be mechanistically accurate.
- The study does not include a full PBPK model to translate in vitro release to in vivo plasma concentrations.

Generalizability

The workflow is generalizable to other drugs and polymer systems, but the specific parameter estimates ($k_{release}$, Weibull parameters) are formulation- and medium-specific. The pH-dependent release mechanism is applicable to other weak bases with pKa in the physiological range.

Key Equations

Higuchi Model

$$Q_t = k_H t^{1/2}$$

Higuchi model for drug release from a matrix system, where Q_t is the cumulative amount released at time t , and k_H is the Higuchi rate constant.

Korsmeyer-Peppas Model

$$\frac{Q_t}{Q_\infty} = k_{KP} t^n$$

Korsmeyer-Peppas model describing drug release from polymeric systems, where Q_t/Q_∞ is the fractional release, k_{KP} is the kinetic constant, and n is the release exponent indicative of the release mechanism.

Weibull Model

$$F(t) = 1 - \exp \left[- \left(\frac{t}{\lambda} \right)^\beta \right]$$

Weibull model for cumulative fractional release, where $F(t)$ is the fraction released at time t , λ is the scale parameter, and β is the shape parameter.

Figures & Tables

- **Figure 1:** Schematic overview of the experimental–modeling–simulation workflow, from microcapsule preparation to population simulation.
 - *Significance:* Provides a clear visual summary of the integrated methodology, highlighting the key steps of data acquisition, kinetic modeling, mechanistic simulation, and scenario analysis.
- **Figure 2:** Comparison of experimental and Weibull model-predicted cumulative release profiles in distilled water (A) and simulated saliva (B).
 - *Significance:* Demonstrates the excellent fit of the Weibull model to the experimental data, with R^2 values of 0.989 and 0.994, confirming its suitability for describing the release kinetics.
- **Figure 3:** Visual Predictive Check (VPC) of the Weibull release model for distilled water (A) and simulated saliva (B), showing 5th, 50th, and 95th percentiles from $N = 200$ simulations.
 - *Significance:* Validates the model’s predictive performance at the population level; all 12 observed data points fall within the 90% prediction interval, supporting model adequacy.
- **Figure 4:** Population-level simulation of sublingual saliva pH variability effects on cumulative diltiazem release, showing profiles by pH group (A) and distribution of cumulative release (B).
 - *Significance:* Illustrates the impact of physiological pH variability on drug release, with lower pH groups showing higher median release, mechanistically linked to the unionized fraction of diltiazem.
- **Figure 5:** Scatter plot showing the inverse relationship between individual saliva pH values and cumulative diltiazem release at 720 min.
 - *Significance:* Provides a clear visualization of the pH-dependent release, reinforcing the mechanistic link and highlighting interindividual variability as a potential source of variable clinical response.
- **Table 1:** Kinetic model fitting results for distilled water and simulated saliva, including model parameters, R^2 , RSS, AIC, and BIC for Higuchi, Korsmeyer-Peppas, and Weibull models.

- *Significance*: Key table for model selection; the Weibull model consistently shows the lowest AIC and BIC values in both media, justifying its use as the reference model for mechanistic simulations.
 - **Table 2**: Summary of Weibull and mechanistic model parameters (Q_{max} , λ , β , $k_{release}$) along with R^2 and RMSE for both media.
 - *Significance*: Provides the final parameter estimates used for simulations and validation, showing the slower release rate in simulated saliva ($k_{release} = 0.00263 \text{ min}^{-1}$) compared to water (0.01112 min^{-1}).
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Code & Reproducibility Assessment

The authors used SimBiology/MATLAB and state that a fixed random seed (rng = 42) was used for reproducibility. No public repository or code sharing is mentioned.

Supplementary Materials

Supplementary materials include a zip file (psp470290-sup-0001-supinfo.zip) containing additional data, likely including Table S1 (raw experimental release data) and Figure S1 (comparison of all three kinetic model fits).

Future Directions

The study raises several important questions and avenues for future research: (1) Validation of the model against independent external in vitro datasets or, ideally, in vivo pharmacokinetic data from animal or human studies. (2) Incorporation of more complex multi-mechanism models (e.g., coupled diffusion-erosion) to better capture the physics of natural polymer microcapsule release. (3) Extension of the pH scenario to include other physiological variables such as saliva volume, flow rate, and enzyme content. (4) Development of a full physiologically based pharmacokinetic (PBPK) model for sublingual absorption to predict plasma concentration-time profiles. (5) Exploration of other weak bases or acids to generalize the pH-dependent release framework.

Expert Commentary

This paper represents a commendable effort to bridge the gap between routine in vitro release testing and mechanistic, predictive modeling—a direction the field should embrace more broadly. The use of AIC/BIC for model selection is rigorous, and the VPC, while internal, is a welcome addition often missing in pharmaceutical science publications. However, the reliance on a first-order two-compartment model for microcapsule release is a simplification; natural polymer systems often exhibit swelling, erosion, and anomalous diffusion that a Weibull model captures empirically but a mechanistic model should ideally reflect. The pH scenario analysis is elegant and clinically relevant, but the assumption that the in vitro release rate scales linearly with the unionized fraction is a strong one. Future work should incorporate multi-mechanism models (e.g., using partial differential equations for diffusion-erosion coupling) and validate against in vivo

data. This study is a solid proof-of-concept that should inspire more integrated experimental-modeling workflows in formulation science.

Bottom Line

This paper presents a well-structured workflow integrating in vitro release data with mechanistic modeling in SimBiology, demonstrating that the Weibull model best describes diltiazem release from gelatin/gum arabic microcapsules. The validated model was extended to simulate the impact of sublingual pH variability, showing that lower pH increases cumulative release due to a higher unionized fraction of diltiazem (pKa 7.7). For practicing pharmacometricians, this study provides a reproducible template for transforming routine dissolution data into predictive, physiologically relevant models, though the internal validation and simplified first-order kinetics limit immediate clinical translation.

☒ Figures

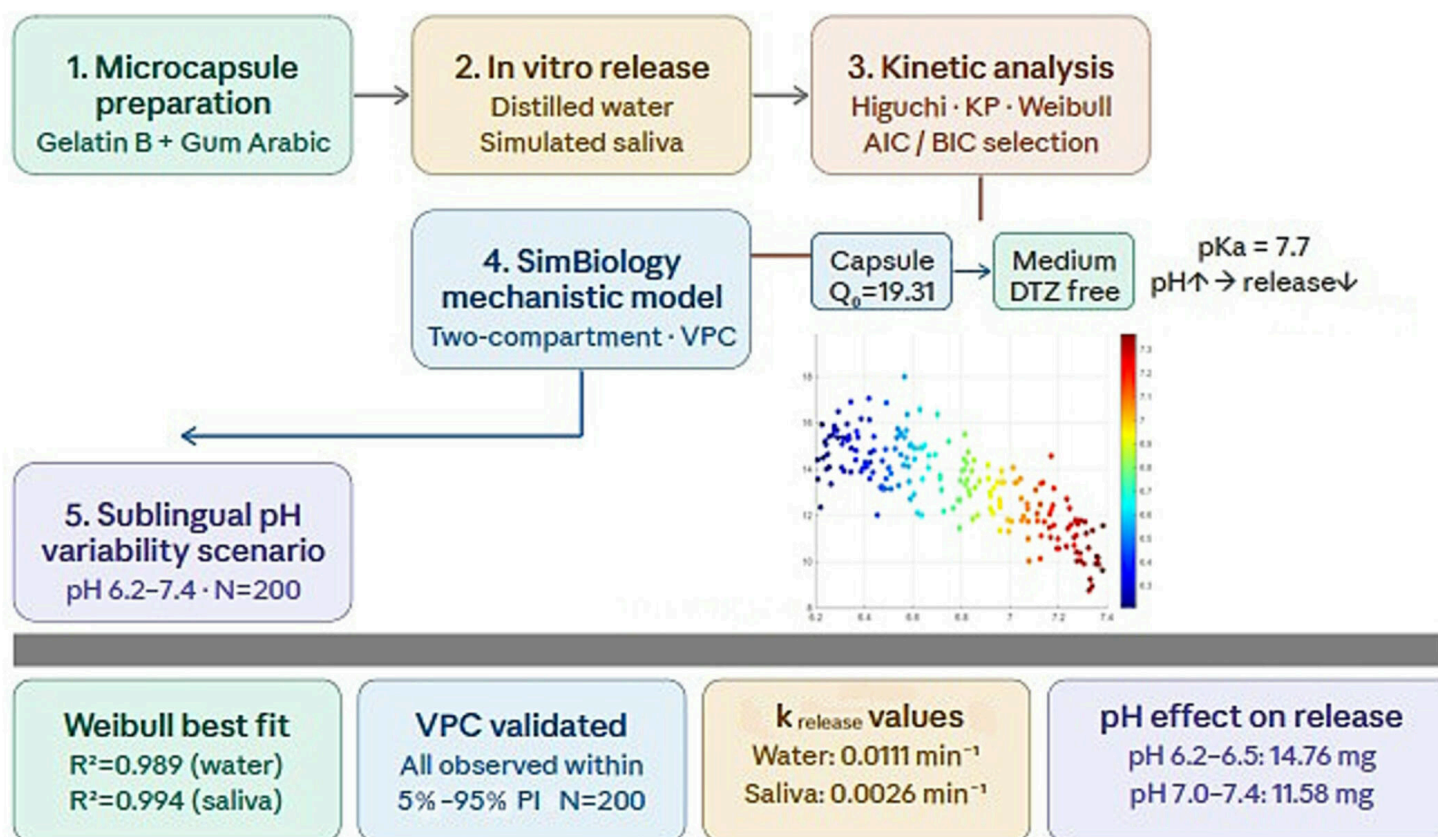


Figure 1: Figure 1

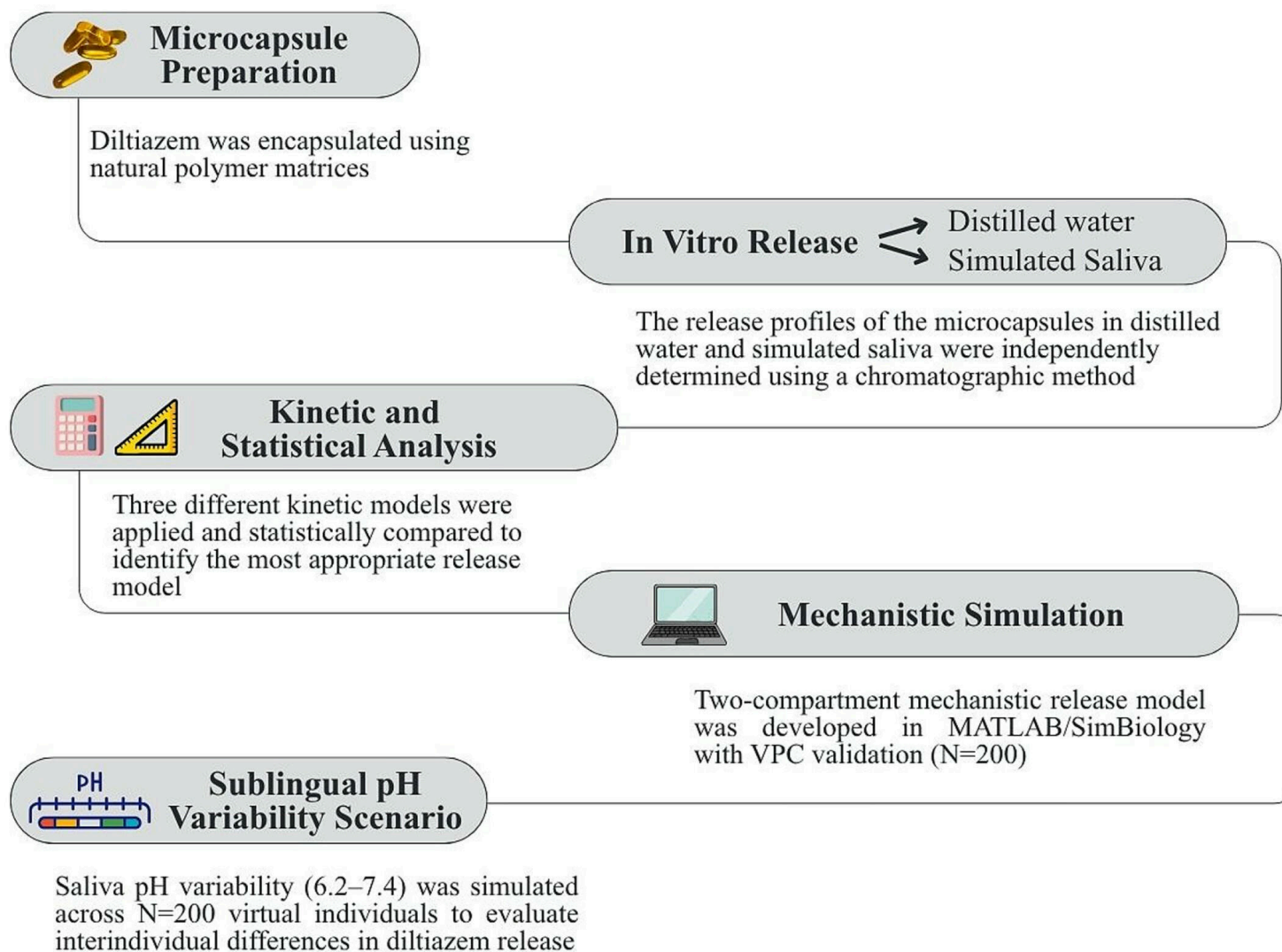


Figure 2: Schematic overview of the experimental-modeling-simulation workflow of the study.

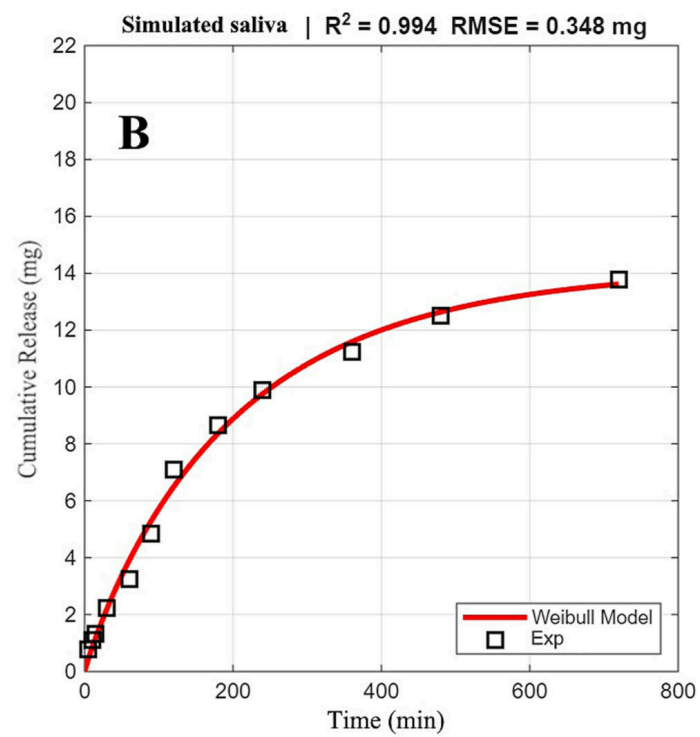
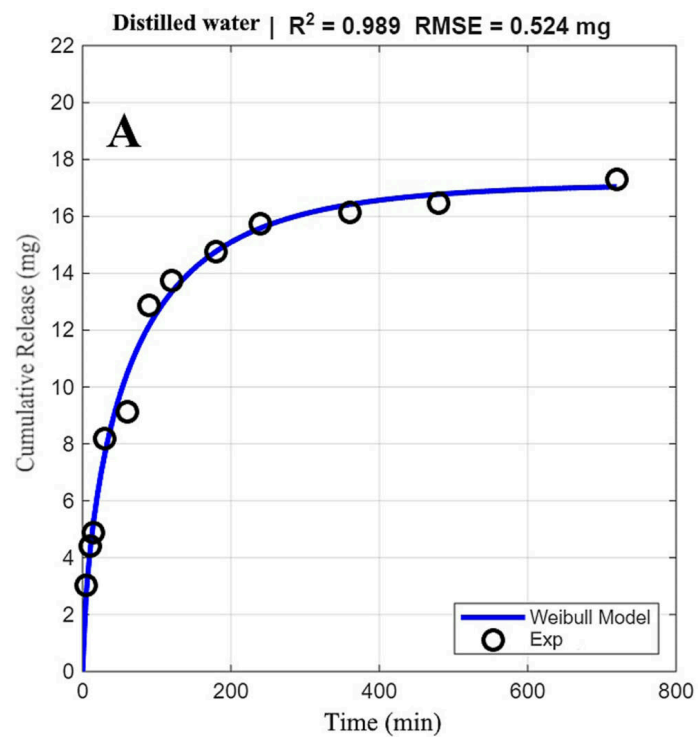


Figure 3: Comparison of experimental and Weibull model-predicted cumulative release profiles of diltiazem in (A) distilled water and (B) simulated saliva.

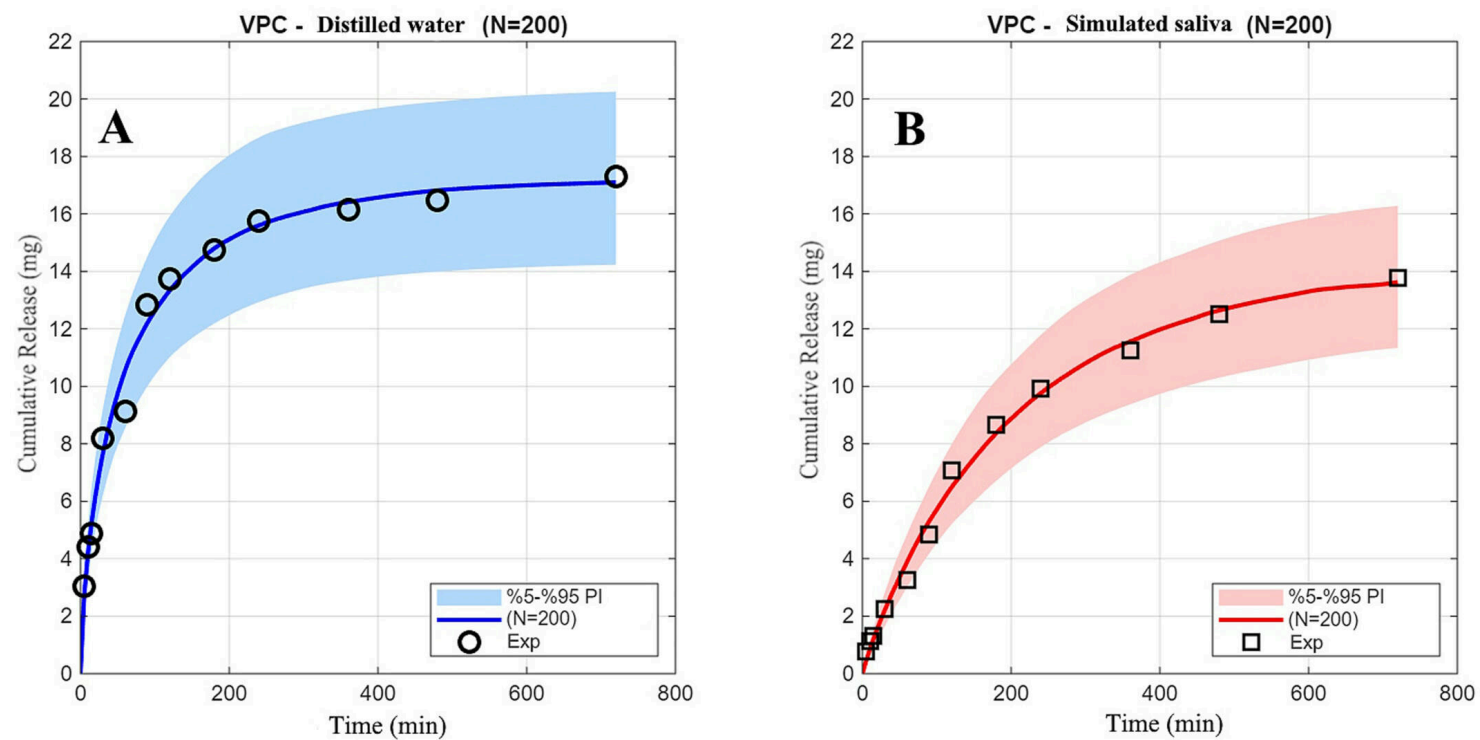


Figure 4: Visual Predictive Check (VPC) of the Weibull release model for (A) distilled water and (B) simulated saliva.

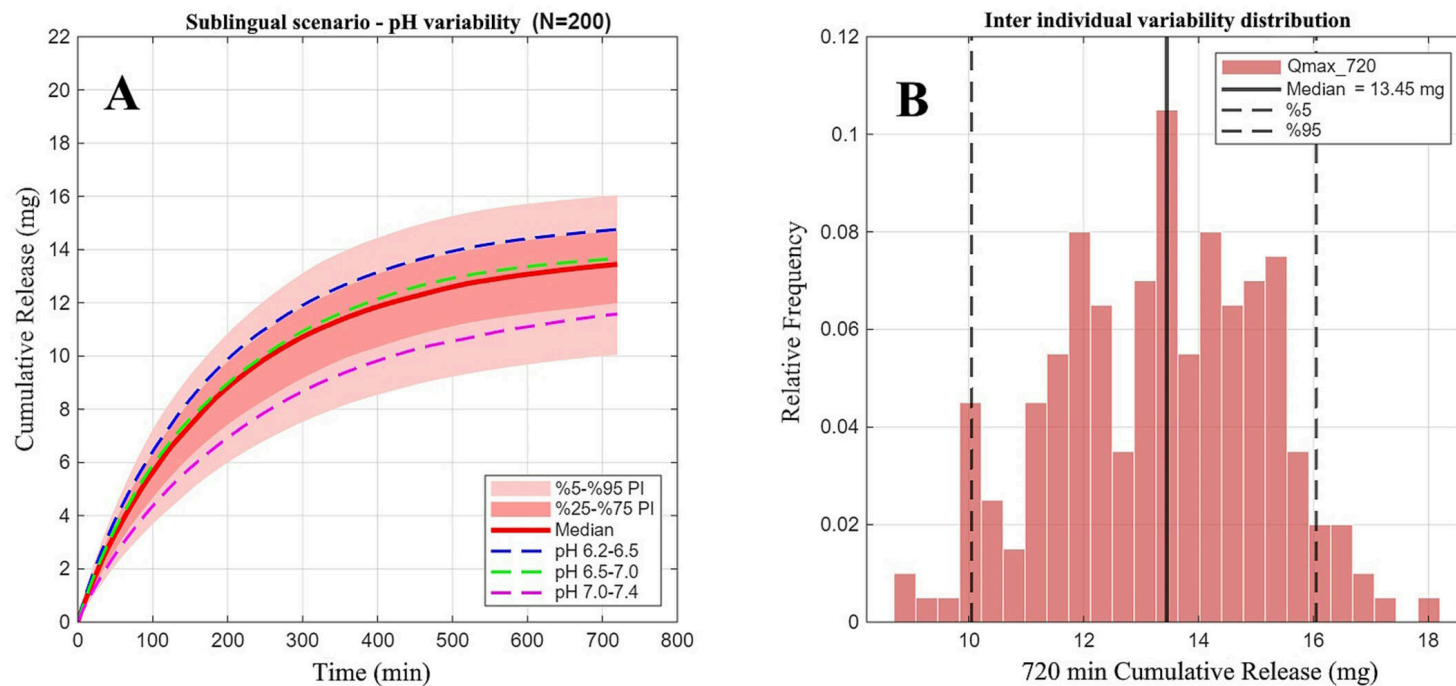


Figure 5: Population-level simulation of sublingual saliva pH variability effects on cumulative diltiazem release. (A) Cumulative release profiles by pH group; (B) distrib

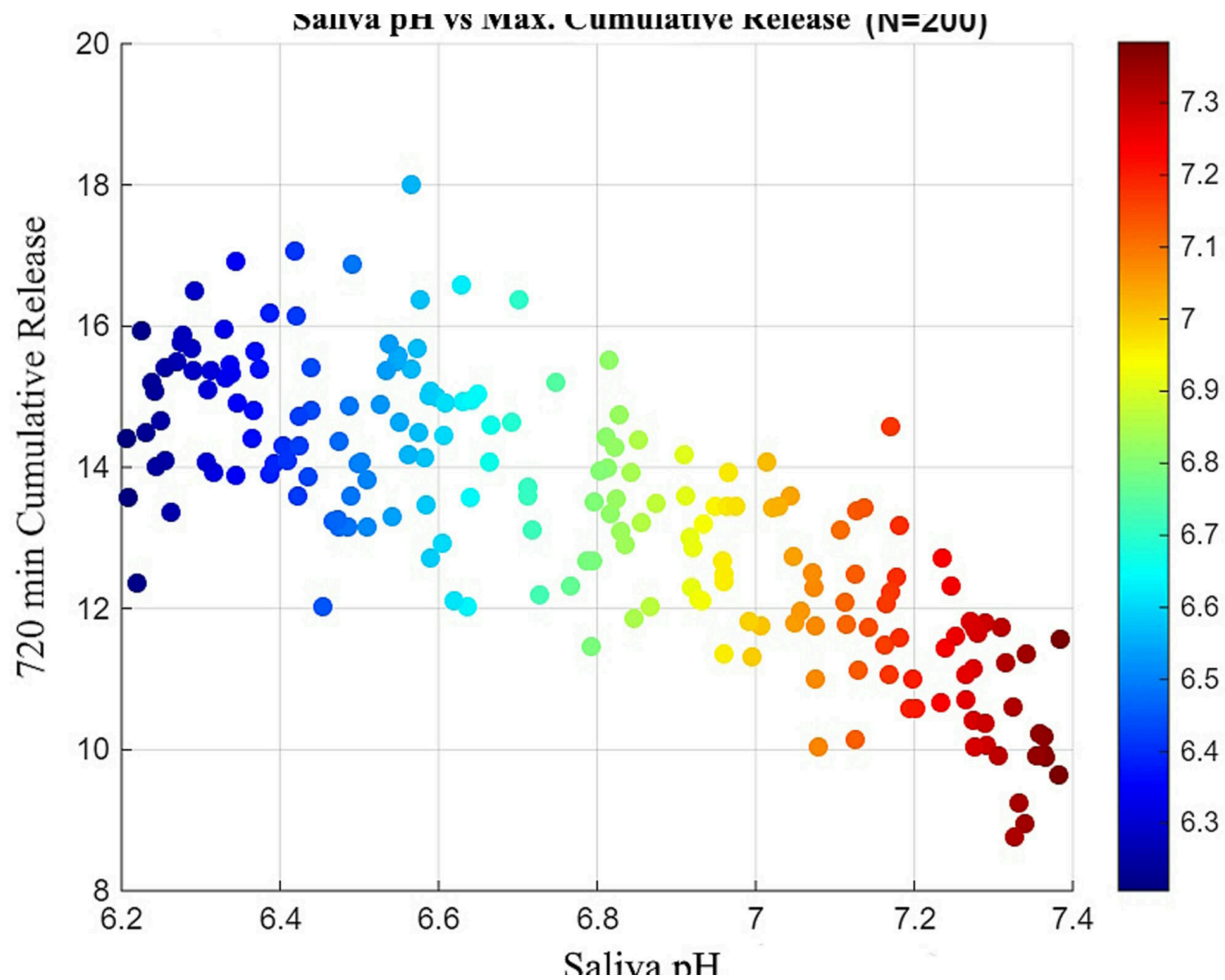


Figure 6: Scatter plot showing the relationship between individual saliva pH values and cumulative diltiazem release at 720 min.